

Qatar-1b

R-2751

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_250621 · created 2026-09-02T23:35:59+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460847.7572 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.151 +/- 0.05
Transit depth (Rp/Rs)^2	2.27 +/- 1.51 [%]
Orbital Inclination (inc)	80.98 +/- 2.99
Airmass coefficient 1 (a1)	1.362 +/- 0.021
Airmass coefficient 2 (a2)	-0.241 +/- 0.07
Scatter in the residuals of the lightcurve fit is	1.55 %
Best Comparison Star	#3 - [49, 216]
Optimal Aperture	2.72
Optimal Annulus	19.3
Transit Duration (day)	0.014 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.151 ± 0.05 vs 0.1463 (deviation 0.09σ)
Duration fit vs published	0.014 d vs 0.0692 d (deviation 4.6σ)
Mid-time O-C	-7.7 min
Depth SNR	3.0
Residual scatter	1.55 %

Light curve

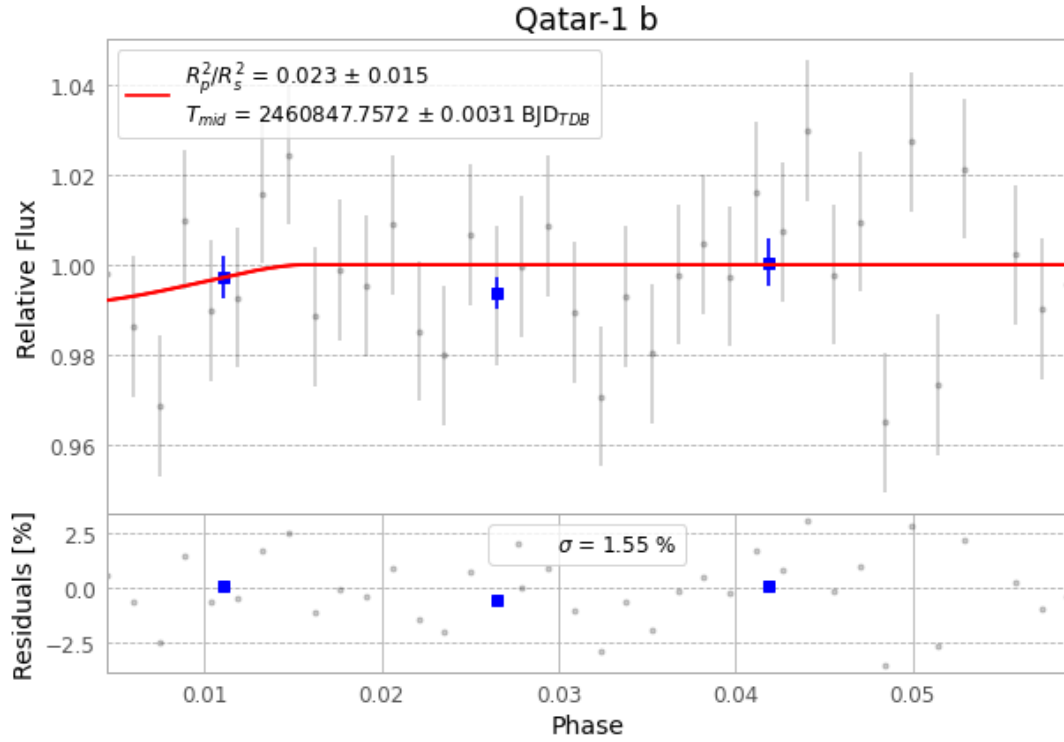


Figure 1 — FinalLightCurve_Qatar-1 b_2025-06-20.png (SHA-256 40cad9912ec1097c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [390, 218], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a54dfe15e0098a70...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

38 FITS frames (25 MB), Qatar-1250621061815.FITS → Qatar-1250621080915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-250621.log (6950e8f5c19f...), FinalLightCurve_Qatar-1 b_2025-06-20.png (40cad9912ec1...), FinalLightCurve_Qatar-1 b_2025-06-20.csv (589e116b9e3a...)

Compute resources

Wall time 151.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-02 23:35Z.